Marmara University
Electrical and Electronics Engineering Department
Spring 2023
EE 4091 - Photonics
Assignment (1) - Due March 24, 2023

The Theory of Light

Part I: Ray optics

1) For the thin prism discussed in class,

   a) Show that

$$\frac{n_{prism}}{n_o} = \frac{\sin[(A + D_{min})/2]}{\sin(A/2)}$$

   where A is the prism angle, n_o and n_{prism} are the refractive indices of the materials outside and inside the prism, respectively, and D_{min} is the minimum deviation angle obtained when the light ray inside the prism becomes parallel to the prism base.

   b) Assuming an air-prism system, plot (using MATLAB) n_{prism} versus D_{min}. Take A=10°.

   c) Using the curve in part (b), predict the prism material (glass, plastic, etc.) for D_{min}=11.5° and D_{min}=15.5°. Find the closest possible material.

Part II: Beam optics

2) Read section 3.1 in the textbook. A 0.2 W semiconductor laser produces a Gaussian Beam at 532 nm wavelength with a spot size of 0.1 mm. Plot the beam diameter and intensity as a function of distance to the surface of the moon (3.5×10^5 km away from earth).
   Comment on your results.

Part III: Wave optics

3) Derive an expression for the average intensity of M identical plane sources with phase shift ϕ between each (same as case in class).

4) Consider the interference pattern of two in-phase monochromatic plane-wave sources with intensities I_1 and I_2 (case studied in class). Plot the interference curve (total intensity) as a function of distance between the sources, for the following cases. (a) The medium is air and the sources are red light, (b) The medium is air and the sources are blue light, (c) The medium is water and the sources are blue light.
   Put all curves on one plot for comparison and give your observations.

Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified